Stakeholder Response

Project Name: Roommate Dashboard

Document ID: StakeholderResponse

Control Label: D0-v1-r1

Date: 4/12/2026

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1. Identified Stakeholders

The following stakeholders were identified during Phase I requirements gathering:

Stakeholder	Type	Role
Roommates (college students, young professionals, anyone sharing housing)	Primary	End users who interact with the application daily to manage bills, chores, and groceries
Household Organizer (a roommate who sets up the household)	Secondary	Sets up the household, generates the invite code, and oversees household activity
Landlord / Property Manager	Secondary	Optional future stakeholder with potential interest in payment accountability features
Application Administrator (project team)	Secondary	Monitors system health and reviews data for debugging purposes

2. How the Team Identified Stakeholder Needs

2.1 Direct Outreach

The team reached out directly to individuals who currently share housing to gather candid feedback on the app concept. One stakeholder was contacted via WhatsApp and asked whether they would see themselves using a roommate dashboard application. Their response confirmed that the core problem was real and that a tool like this would be genuinely useful:

> *"Yh pretty solid idea. Makes life pretty easy with the roommates to be honest. Sometimes I be forgetting what payments I owe to my roommates. Something like that would actually be super useful."*

This direct feedback validated the primary pain point — roommates losing track of shared financial obligations — and informed the decision to prioritize the expense tracking and payment logging features in the implementation.

2.2 Structured Stakeholder Analysis

Beyond direct outreach, the team performed a structured stakeholder analysis during Phase I. Each stakeholder group's needs were documented and organized by category:

Stakeholder	Identified Needs
Roommates	Fair bill splitting, visibility of who has paid and who has not, fair chore assignment with rotation, shared grocery list to prevent double-buying
Household Organizer	Easy setup process, ability to invite roommates quickly via invite code, dashboard view of full household status
Administrator	Basic system monitoring, ability to review data for debugging purposes

2.3 Requirements Elicitation

The identified needs were translated into formal requirements through elicitation sessions documented in ElicitationRequirements_D0-v1-r1.pdf. User stories and action diagrams (UserStories_ActionDiagrams_D0-v1-r1.pdf) were then created to capture stakeholder workflows, ensuring that every feature in the application traced back to a concrete stakeholder need.

3. How the Final Vision and Implementation Satisfied Stakeholder Requirements

3.1 Roommate Needs

Stakeholder Need	How It Was Satisfied
Fair bill splitting	The Expense module supports both equal and custom percentage splits. ExpenseService calculates each roommate's share automatically and stores it as an ExpenseSplit record.
Visibility of who has paid and who has not	Each ExpenseSplit has a paid flag and paidAt timestamp. The dashboard surfaces unpaid splits so all roommates can see outstanding balances at a glance.

Stakeholder Need	How It Was Satisfied
Fair chore assignment with rotation	The Chore module assigns chores to specific roommates and supports a rotating flag. ChoreService.rotateWeeklyChores() automatically cycles assignments across all household members.
Shared grocery list, prevent double-buying	The GroceryItem module maintains a household-scoped list with NEEDED and PURCHASED statuses. Once an item is marked purchased it is visually distinguished, preventing duplication.

3.2 Household Organizer Needs

Stakeholder Need	How It Was Satisfied
Easy setup process	HouseholdService.createHousehold() provisions a new household in a single form submission. The organizer is automatically assigned the ORGANIZER role via HouseholdMember.
Ability to invite roommates quickly	A unique 8-character alphanumeric invite code is generated automatically for every household. Roommates join by entering the code on the household-join page.
Dashboard view of household status	The main dashboard aggregates open chores, unpaid expense splits, recent activity logs, and grocery items in a single view assembled by DashboardSummaryService and DashboardModelBuilder.

3.3 Administrator Needs

Stakeholder Need	How It Was Satisfied
Basic system monitoring	Spring Boot Actuator endpoints are available for health and status monitoring. The application logs errors through GlobalExceptionHandler, providing visibility into system issues.
Data review for debugging purposes	The ActivityLog entity maintains an immutable audit trail of all domain events (expenses added, chores completed, payments logged, households created/joined). Each log entry records the actor, action type, message, and timestamp.

4. Summary

Every stakeholder need identified in Phase I was carried forward into the Phase II implementation. The direct outreach conversation confirmed the core value proposition early, which guided the team to prioritize expense tracking and payment visibility as the most critical features. The structured requirements process — elicitation, user stories, and use case diagrams — ensured that chore rotation, grocery management, and household setup were also fully implemented and traceable to documented stakeholder needs. The final Roommate Dashboard application satisfies all identified primary and secondary stakeholder requirements as documented in this phase.